Planning and Approximate Reasoning (PAR)

Final Practical Project

Autonomous Drone Planning for Smart-City Emergency Response

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1 Introduction

After an emergency in a city, a fleet of autonomous drones has to bring supplies such as medical kits, water and communication equipment from depots to the affected locations. Each drone carries one item at a time, and its battery limits how far it can fly before it needs a charging station. Some areas are blocked at first and only become reachable once a nearby site has been inspected or has received its supply. The goal is to find a sequence of flights, inspections, pick-ups, deliveries and recharges that meets every demand at the lowest total cost.

In this project we model the scenario in PDDL and solve it with Fast Downward. Since the planner does not support numeric fluents, the battery is encoded with symbolic levels. We build two alternative models that differ only in how recharging works: a full recharge in one action (Model A), or one level per action (Model B). We test both on a small, a medium and a large instance with several planner configurations, validate every plan with VAL, and compare plan quality and search effort.

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